# Clinical Balance Assessment

Prof Isaac Sorinola

PT7003

# Learning Outcomes

**By the end of the session the student should be able to:**

- Describe components included in a thorough balance assessment
- Explain the differences between static and dynamic balance tests
- Explain the difference between a functional and an impairment-based assessment
- Discuss the issues of reliability and validity as they relate to a balance assessment



# Why assess balance?

- Balance deficits are common in neurologically impaired patients
- Common impairments are:
- **Motor dysfunction:**
  - Musculoskeletal: ↓ strength; trunk instability; ↓ range of movement; altered muscle tone
  - Biomechanics: ↓ LOS; ↓ balance response; altered movement strategies
- **Sensory dysfunction:**
  - Altered sensations:↓ ankle proprioception; vestibular damage; dizziness; visual deficits
  - Altered sensory integration: impaired attention; impaired sensory weighting; delayed anticipatory postural response; altered perceived stability limits; abnormal perception of vertical.

# Balance Assessment

1. Patient safety
2. Quantifying functional limitations
3. Identifying underlying impairments



# Patient safety





Close guarding of patients always

# Balance Assessment

1. Patient safety
2. Quantifying functional limitations
3. Identifying underlying impairments



# Clinical Balance Assessment - Task

- Balance assessment should be performed in the context of a functional activity – bed mobility, sitting, standing, walking

- Set an appropriate balance task
  - Safe situation (environment, assistance)
  - Incorporate various environmental contexts
  - Challenge patient balance:
    - reduce base of support, raising COG, displacing COG toward LOS, unstable surface, altering sensory input
  - Assess all aspects of balance responses:
    - posture & movement, balance strategies, balance reactions, all planes of movement

# Clinical Balance Assessment - Observation

- Can they balance? Posture, excessive sway
- Can they balance during movement? Strategies used, are they appropriate?
- Does the patient use compensatory strategies
- Is the goal completed successfully?
- Is the task effortful?
- What happens to the balance if sensory input is reduced? (eyes closed)

# Assessment of balance: Quantifying functional limitations

Commonly used Outcome measures can:

- Determine performance level in functional balance tasks
- Indicate need for therapy
- Serve as a baseline level of performance
- Provide objective documentation of functional status
- To determine effectiveness of therapy programmes

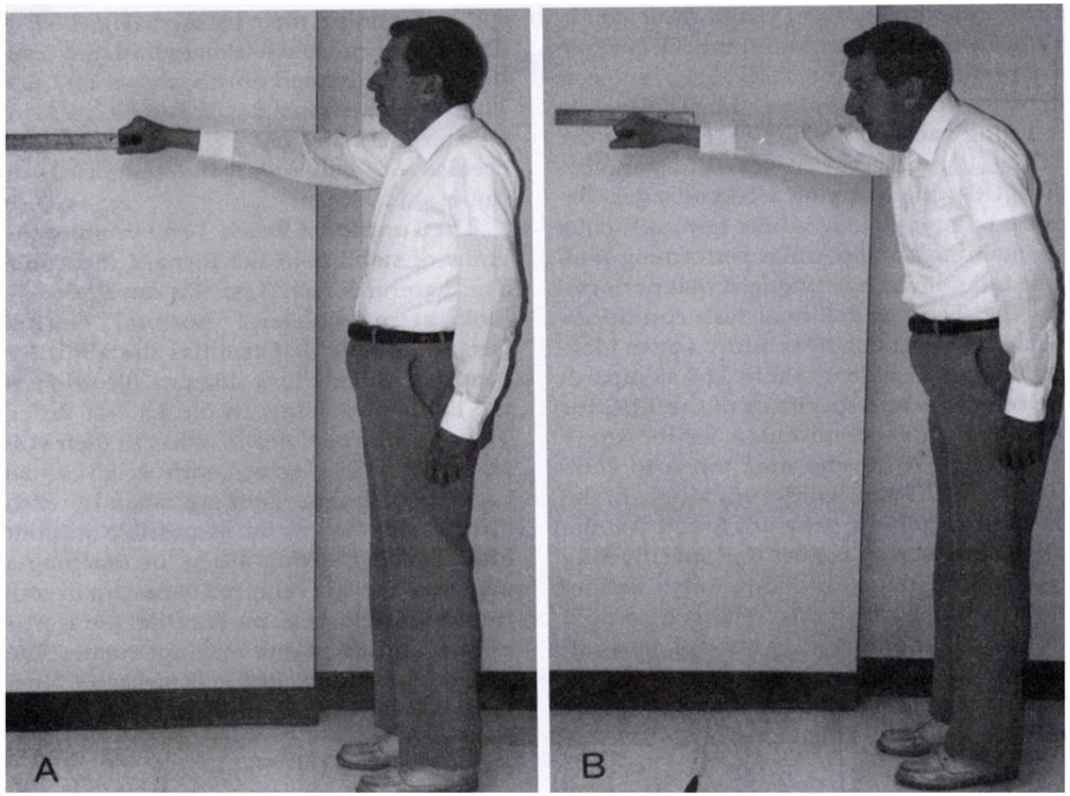
# Quantifying functional limitations: Get up and Go test (Timed up and Go (TUG) test)

- quick screening tool for detecting imbalance in the elderly (Mathias S et al, 1986)
- reliable and consistent
- correlates well with functional capacity
- highly correlated with gait velocity (Evans and Krebs, 1999)



# Quantifying functional limitations: Functional Reach Test

- developed to detect balance problems in older adults (Duncan et al., 1990)
- tests limits of stability in forward direction
- variable findings: reliable, highly predictive of falls, unable to differentiate fallers / non-fallers
- does NOT test dynamic balance (Wernick-Robinson et al., 1999)




**Scores < 6 or 7 inches = limited functional balance;**
**>10 inches = normal values**

# Quantifying functional limitations: Dynamic Gait Index

- assesses ability to modify gait according to changing task demands (Shumway-Cook and Woollacott, 1995)
- valid predictor of falls in both older adults (Shumway-Cook, 1997) and vestibular patients (Whitney et al 2000)
- Functional Gait Index (Wrisley et al., 2004)
- Clinically meaningful change



# Quantifying functional limitations: Tinetti Performance & Mobility Assessment

- Developed to assess balance & mobility in older adults (Tinetti, 1986)
- Performance-Oriented Mobility Assessment (POMA)
- Assesses likelihood of falls
- Used often with gait evaluation
  - assesses different functions than DGI
- Good inter-rater reliability

- Armless chair, stopwatch and 15feet walkway (4.5metres)
- Patient stands from sitting & stay standing
- Turns 360 degrees and sit back down (Balance)
- Patient walks a few meters at normal speed, followed by turning and walking back at a fast but safe speed.
- The patient then sits back down (mobility)
- The assessor grades the patient's performance.
- Uses a 3-point ordinal scale (0-2)
- Gait (scored over 12 and balance over 16 points = 28 points

# Quantifying functional limitations: Berg Balance Scale

- Used to objectively determine a patient's ability to safely balance during a series of pre-determined tasks
- 14 item list
- Assessed on a 5-point ordinal scale (0-4)
- Total: 56

- Good re-test and inter-rater reliability
- Effective predictor of falls in community-dwelling adults (Shumway-Cook et al., 1997)
- Suggested better at identifying individuals w/ greater impairment vs less impaired w/ risk for falls

https://youtu.be/i_Jlc5kL6KU

# Berg Balance Scale (with instructions)

SITTING TO STANDING
INSTRUCTIONS: Please stand up. Try not to use your hand for support.
( ) 4 able to stand without using hands and stabilize independently
( ) 3 able to stand independently using hands
( ) 2 able to stand using hands after several tries
( ) 1 needs minimal aid to stand or stabilize
( ) 0 needs moderate or maximal assist to stand

STANDING UNSUPPORTED
INSTRUCTIONS: Please stand for two minutes without holding on.
( ) 4 able to stand safely for 2 minutes
( ) 3 able to stand 2 minutes with supervision
( ) 2 able to stand 30 seconds unsupported
( ) 1 needs several tries to stand 30 seconds unsupported
( ) 0 unable to stand 30 seconds unsupported

If a subject is able to stand 2 minutes unsupported, score full points for sitting unsupported. Proceed to item #4.

SITTING WITH BACK UNSUPPORTED BUT FEET SUPPORTED ON FLOOR OR ON A STOOL
INSTRUCTIONS: Please sit with arms folded for 2 minutes.
( ) 4 able to sit safely and securely for 2 minutes
( ) 3 able to sit 2 minutes under supervision
( ) 2 able to able to sit 30 seconds
( ) 1 able to sit 10 seconds
( ) 0 unable to sit without support 10 seconds

STANDING TO SITTING
INSTRUCTIONS: Please sit down.
( ) 4 sits safely with minimal use of hands
( ) 3 controls descent by using hands
( ) 2 uses back of legs against chair to control descent
( ) 1 sits independently but has uncontrolled descent
( ) 0 needs assist to sit

TRANSFERS
INSTRUCTIONS: Arrange chair(s) for pivot transfer. Ask subject to transfer one way toward a seat with armrests and one way toward a seat without armrests. You may use two chairs (one with and one without armrests) or a bed and a chair.
( ) 4 able to transfer safely with minor use of hands
( ) 3 able to transfer safely definite need of hands
( ) 2 able to transfer with verbal cuing and/or supervision
( ) 1 needs one person to assist
( ) 0 needs two people to assist or supervise to be safe

STANDING UNSUPPORTED WITH EYES CLOSED
INSTRUCTIONS: Please close your eyes and stand still for 10 seconds.
( ) 4 able to stand 10 seconds safely
( ) 3 able to stand 10 seconds with supervision
( ) 2 able to stand 3 seconds
( ) 1 unable to keep eyes closed 3 seconds but stays safely
( ) 0 needs help to keep from falling

STANDING UNSUPPORTED WITH FEET TOGETHER
INSTRUCTIONS: Place your feet together and stand without holding on.
( ) 4 able to place feet together independently and stand 1 minute safely
( ) 3 able to place feet together independently and stand 1 minute with supervision
( ) 2 able to place feet together independently but unable to hold for 30 seconds

**Berg Balance Scale**

( ) 1 needs help to attain position but able to stand 15 seconds feet together
( ) 0 needs help to attain position and unable to hold for 15 seconds

REACHING FORWARD WITH OUTSTRETCHED ARM WHILE STANDING
INSTRUCTIONS: Lift arm to 90 degrees. Stretch out your fingers and reach forward as far as you can. (Examiner places a ruler at the end of fingertips when arm is at 90 degrees. Fingers should not touch the ruler while reaching forward. The recorded measure is the distance forward that the fingers reach while the subject is in the most forward lean position. When possible, ask subject to use both arms when reaching to avoid rotation of the trunk.)
( ) 4 can reach forward confidently 25 cm (10 inches)
( ) 3 can reach forward 12 cm (5 inches)
( ) 2 can reach forward 5 cm (2 inches)
( ) 1 reaches forward but needs supervision
( ) 0 loses balance while trying/requires external support

PICK UP OBJECT FROM THE FLOOR FROM A STANDING POSITION
INSTRUCTIONS: Pick up the shoe/slipper, which is place in front of your feet.
( ) 4 able to pick up slipper safely and easily
( ) 3 able to pick up slipper but needs supervision
( ) 2 unable to pick up but reaches 2-5 cm(1-2 inches) from slipper and keeps balance independently
( ) 1 unable to pick up and needs supervision while trying
( ) 0 unable to try/needs assist to keep from losing balance or falling

TURNING TO LOOK BEHIND OVER LEFT AND RIGHT SHOULDERS WHILE STANDING
INSTRUCTIONS: Turn to look directly behind you over toward the left shoulder. Repeat to the right. Examiner may pick an object to look at directly behind the subject to encourage a better twist turn.
( ) 4 looks behind from both sides and weight shifts well
( ) 3 looks behind one side only other side shows less weight shift
( ) 2 turns sideways only but maintains balance
( ) 1 needs supervision when turning
( ) 0 needs assist to keep from losing balance or falling

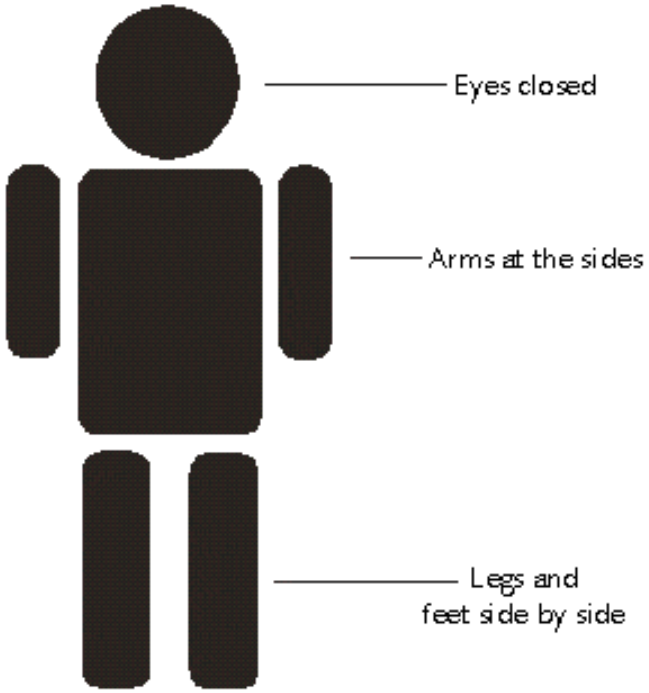
TURN 360 DEGREES
INSTRUCTIONS: Turn completely around in a full circle. Pause. Then turn a full circle in the other direction.
( ) 4 able to turn 360 degrees safely in 4 seconds or less
( ) 3 able to turn 360 degrees safely one side only 4 seconds or less
( ) 2 able to turn 360 degrees safely but slowly
( ) 1 needs close supervision or verbal cuing
( ) 0 needs assistance while turning

PLACE ALTERNATE FOOT ON STEP OR STOOL WHILE STANDING UNSUPPORTED
INSTRUCTIONS: Place each foot alternately on the step/stool. Continue until each foot has touch the step/stool four times.
( ) 4 able to stand independently and safely and complete 8 steps in 20 seconds
( ) 3 able to stand independently and complete 8 steps in > 20 seconds
( ) 2 able to complete 4 steps without aid with supervision
( ) 1 able to complete > 2 steps needs minimal assist
( ) 0 needs assistance to keep from falling/unable to try

STANDING UNSUPPORTED ONE FOOT IN FRONT
INSTRUCTIONS: (DEMONSTRATE TO SUBJECT) Place one foot directly in front of the other. If you feel that you cannot place your foot directly in front, try to step far enough ahead that the heel of your forward foot is ahead of the toes of the other foot. (To score 3 points, the length of the step should exceed the length of the other foot and the width of the stance should approximate the subject's normal stride width.)
( ) 4 able to place foot tandem independently and hold 30 seconds
( ) 3 able to place foot ahead independently and hold 30 seconds
( ) 2 able to take small step independently and hold 30 seconds
( ) 1 needs help to step but can hold 15 seconds
( ) 0 loses balance while stepping or standing

STANDING ON ONE LEG

# Quantifying functional limitations: Romberg




- First described in 1846
- Appropriate test to diagnose sensory ataxia – gait problems caused by abnormal proprioception
- determines if balance issues are related to the function of the dorsal column

# Quantifying functional limitations: Single leg Stance

- This is used to assess static postural and balance control
- Performed with eyes open and hands on the hips
- Patient stands unassisted on one leg, timed from the time the other foot leaves the ground till when the foot touches the ground again or the arm leaves the hips
- If unable to stand for 5 seconds or less, patient is at greater risk of injury from fall.



# Quantifying functional limitations: Commonly used tests: 'Walking while talking (WWT) test' (Vorghese et al., 2002)

- This is a dual-task measure of divided attention to examine cognitive –motor interactions, especially for identifying fallers.
- Patient walks 6m, turn and return (12m).
- Patient is instructed to:
  - WWT simple: walking while reciting the alphabet loud
  - WWT complex: walking while reciting alternate letters of the alphabet aloud.
- Scoring: time needed to complete the distance

- **Simple:** Walk and recite letters of the alphabet; High specificity (89.4%) for identifying falls if completed in 20 seconds or longer
- **Complex:** Walk and recite alternate letters of the alphabet; High specificity (95.6%) for identifying falls if completed in 33 seconds or longer

# Balance Assessment

1. Patient safety
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# Balance Assessment

Involves:

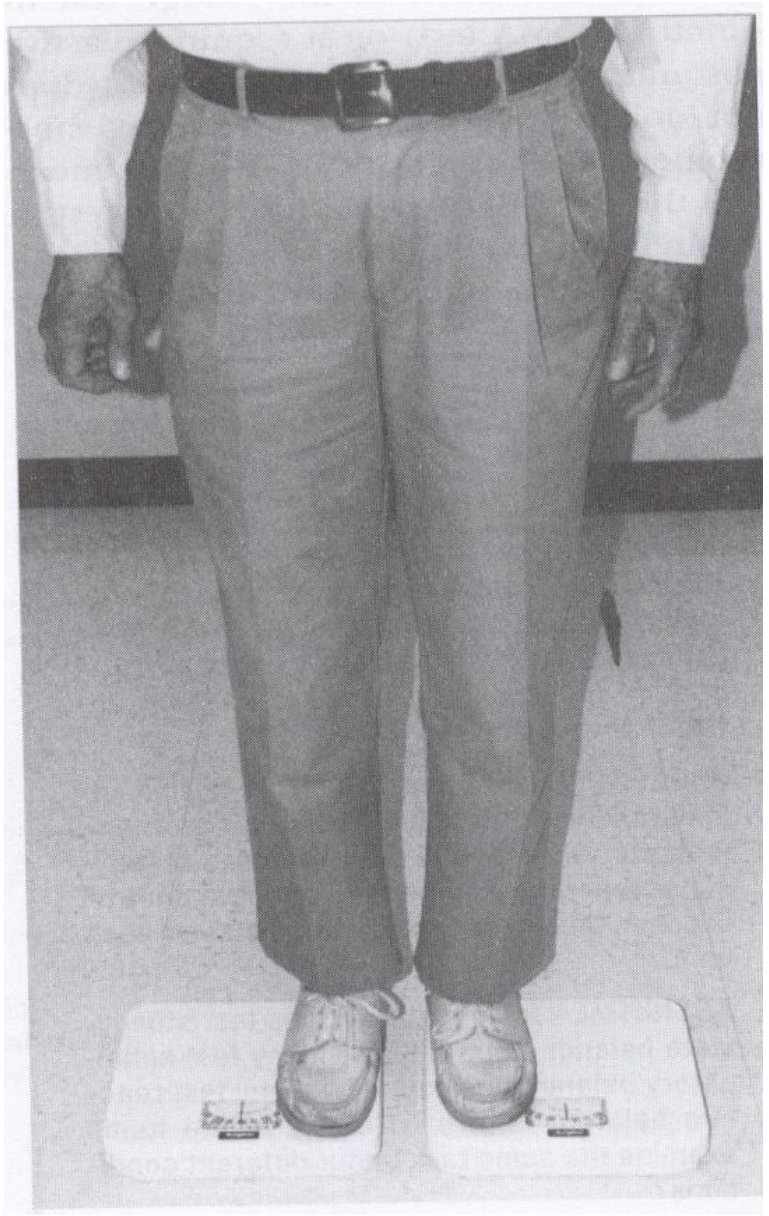
1. Patient safety
2. Quantifying functional limitations
3. Identifying underlying impairments
    A. Musculoskeletal
    B. Neuromuscular
    C. Postural orientation

# Identifying underlying impairments

A. Musculoskeletal impairments:
   - Range of Movement (ROM)
   - Muscle strength
   - Pain
   - Internal perception of vertical - postural alignment

# Identifying underlying impairments

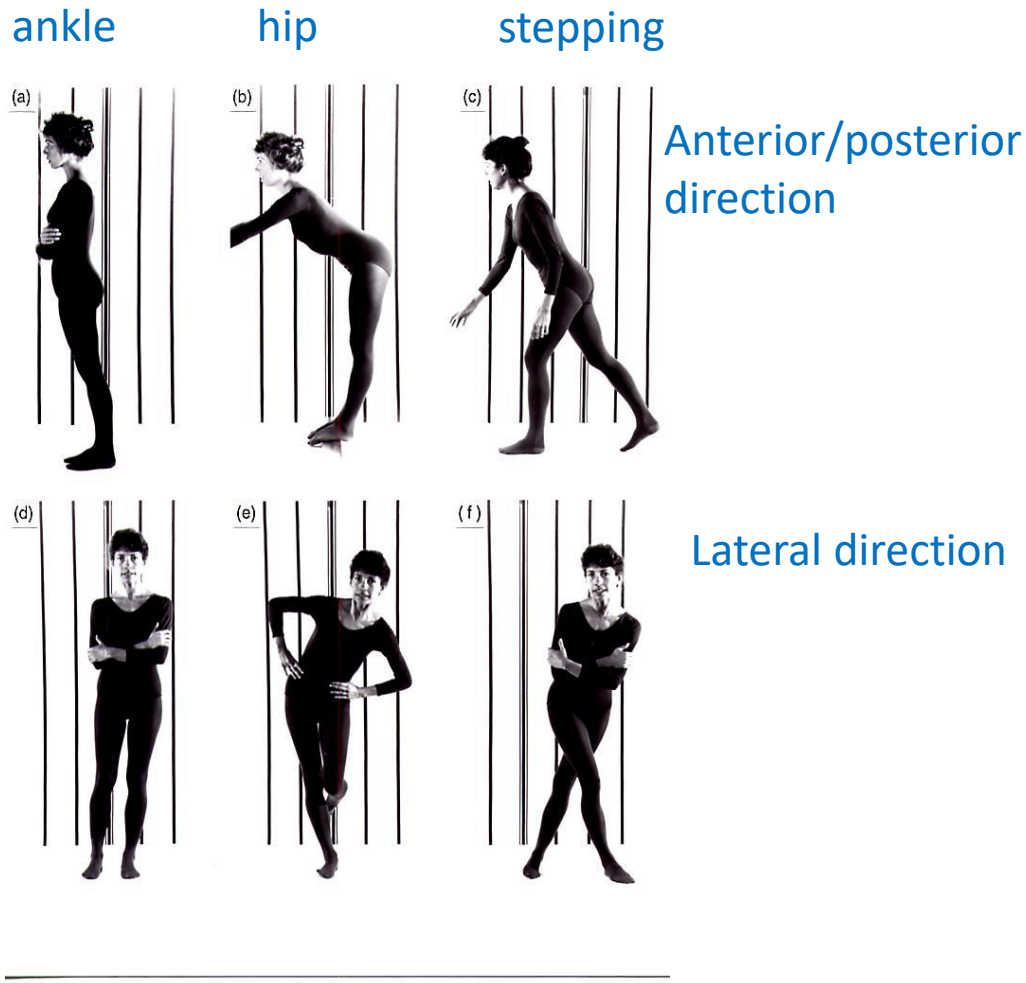
- A: Musculoskeletal impairments
- Internal perception of vertical - postural alignment



# Identifying underlying impairments

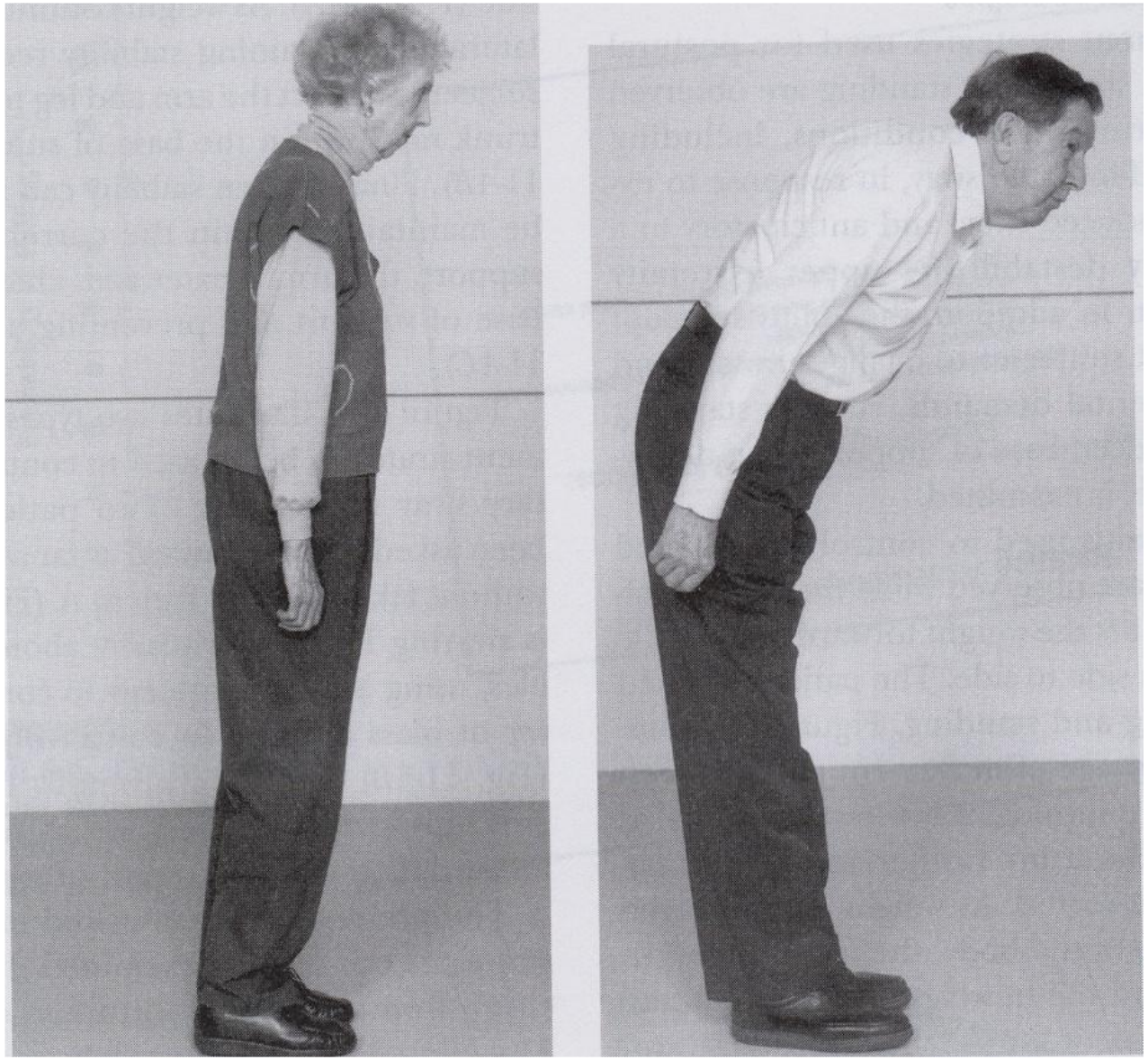
B: Neuromuscular Impairments

May affect ability to generate and co-ordinate multi-joint movement necessary for balance



# Assessing of balance: identifying underlying impairments

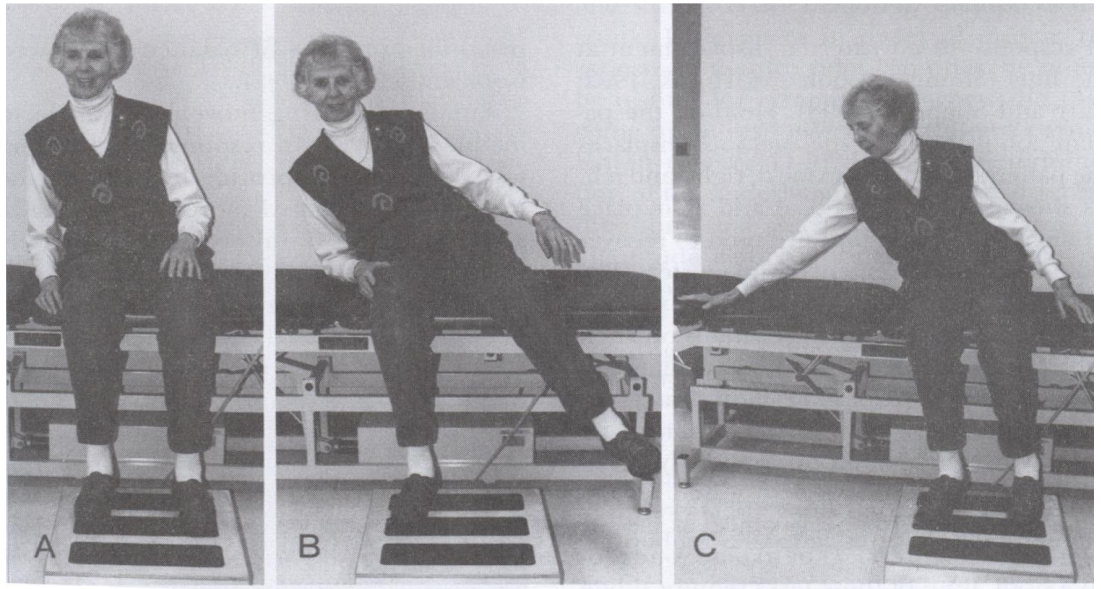
B. Neuromuscular impairments
   - Assessment: Postural movement strategies examined under 3 different task conditions
     - Self-initiated postural sway



# Assessing of balance: identifying underlying impairments

## B. Neuromuscular impairments

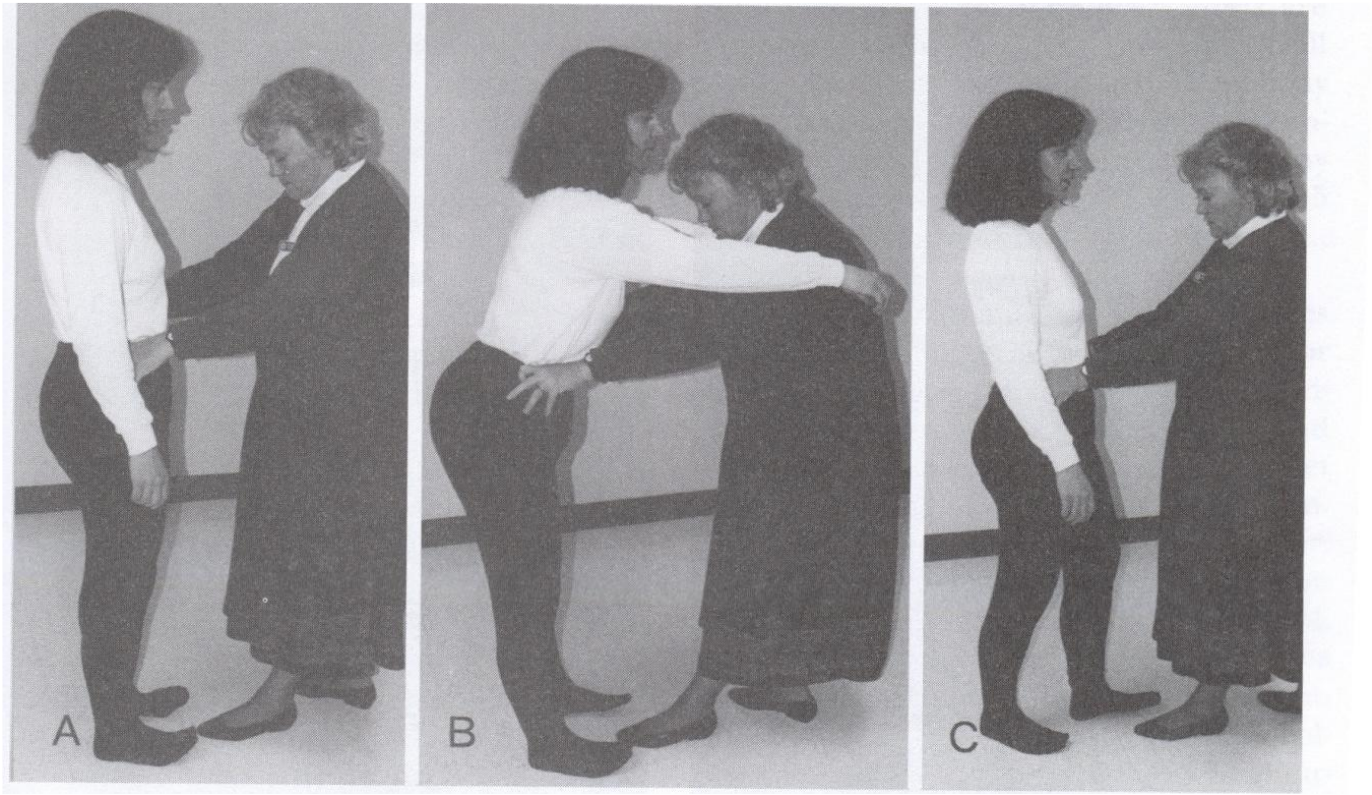
- Assessment: Postural movement strategies examined under 3 different task conditions
  - Self-initiated postural sway



# Assessing of balance: identifying underlying impairments

B. Neuromuscular impairments

- Assessment: Postural movement strategies examined under 3 different task conditions
  - externally induced postural adjustments



# Assessing of balance: identifying underlying impairments

- Neuromuscular impairments

- Assessment: Postural movement strategies examined under 3 different task conditions
  - anticipatory postural adjustments accompanying a potentially de-stabilising limb movement



# Assessing of balance: Identifying underlying impairments

C. Postural orientation

Disruption of sensory information processing may affect ability to:

- Orientate body with respect to gravity
- Adapt and re-weigh sensory inputs according to changes in task and environmental demands
- Anticipate instability based on prior experience

# Assessing of balance: identifying underlying impairments

C. Postural orientation

Assessment:

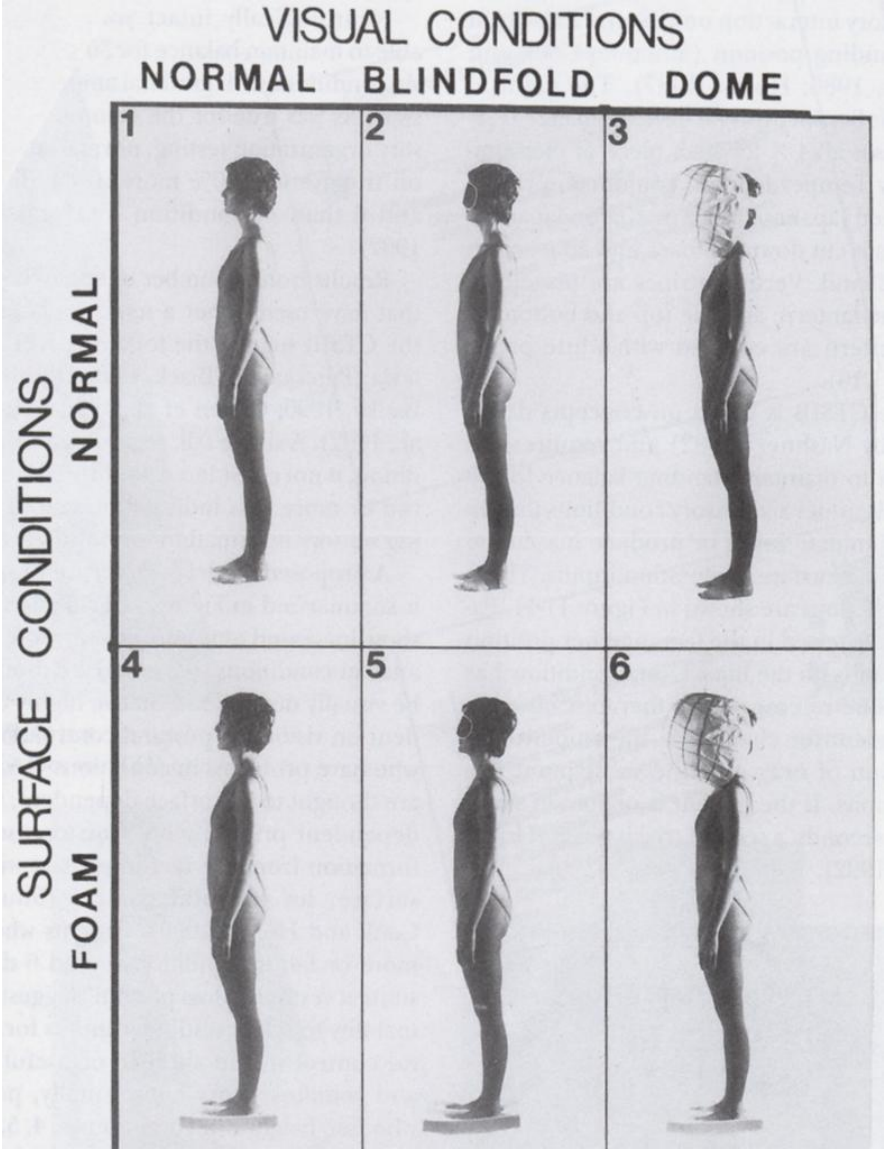
- Proprioception: vibration, joint position sense
- Sensory integration:
  - Dynamic computerized posturography,
  - Clinical Test for Sensory Integration for Balance (CTSIB):

# Assessing of balance: identifying underlying impairments

- Postural orientation

<u>Assessment:</u>

- Sensory integration:
  - Dynamic computerized posturography,
  - Clinical Test for Sensory Integration for Balance (CTSIB)



# BESTest (Horak et al 2009) & Mini BESTest (Franchignoni et al., 2010)

- A tool developed to analyse several postural control systems that may contribute to poor functional balance in adults; Mini BESTest (Franchignoni et al., 2010)
- BESTest: Score of 69% cut-off for identifying fallers in people with Parkinson's Disease (Leddy et al., 2011); 92% accuracy for identifying fallers in people with MS (Jacobs and Kasser, 2012)
- Accuracy of identifying a faller comparable for the Mini-BESTest and BESTest (PD) (Leddy et al., 2011b)
- Mini-BESTest: able to classify individual patients who show significant improvement in balance function; Minimal important change 4 points (Godi et al., 2013).

It has 36 items that evaluate performance of 6 balance systems:

1. Bio-mechanical constraints
2. Stability limits/verticality
3. Anticipatory responses
4. Postural responses
5. Sensory orientation
6. Stability in gait

www.bestest.us/files/4413/6358/0759/BESTest.pdf

**Mini-BESTest:** Balance Evaluation Systems Test
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***ANTICIPATORY*** ***SUB SCORE:*** /6

**1. SIT TO STAND**
*Instruction: "Cross your arms across your chest. Try not to use your hands unless you must. Do not let your legs lean against the back of the chair when you stand. Please stand up now."*
(2) Normal: Comes to stand without use of hands and stabilizes independently.
(1) Moderate: Comes to stand WITH use of hands on first attempt.
(0) Severe: Unable to stand up from chair without assistance, OR needs several attempts with use of hands.

**2. RISE TO TOES**
*Instruction: "Place your feet shoulder width apart. Place your hands on your hips. Try to rise as high as you can onto your toes. I will count out loud to 3 seconds. Try to hold this pose for at least 3 seconds. Look straight ahead. Rise now."*
(2) Normal: Stable for 3 s with maximum height.
(1) Moderate: Heels up, but not full range (smaller than when holding hands), OR noticeable instability for 3 s.
(0) Severe: ≤ 3 s.

**3. STAND ON ONE LEG**
*Instruction: "Look straight ahead. Keep your hands on your hips. Lift your leg off of the ground behind you without touching or resting your raised leg upon your other standing leg. Stay standing on one leg as long as you can. Look straight ahead. Lift now."*

***Left:*** *Time in Seconds Trial 1:_____Trial 2:_____*
(2) Normal: 20 s.
(1) Moderate: < 20 s.
(0) Severe: Unable.

***Right:*** *Time in Seconds Trial 1:_____Trial 2:_____*
(2) Normal: 20 s.
(1) Moderate: < 20 s.
(0) Severe: Unable

**To score each side separately use the trial with the longest time.**
**To calculate the sub-score and total score use the side [left or right] with the lowest numerical score [i.e. the worse side].**

***REACTIVE POSTURAL CONTROL*** ***SUB SCORE:*** /6

**4. COMPENSATORY STEPPING CORRECTION- FORWARD**
*Instruction: "Stand with your feet shoulder width apart, arms at your sides. Lean forward against my hands beyond your forward limits. When I let go, do whatever is necessary, including taking a step, to avoid a fall."*
(2) Normal: Recovers independently with a single, large step (second realignment step is allowed).
(1) Moderate: More than one step used to recover equilibrium.
(0) Severe: No step, OR would fall if not caught, OR falls spontaneously.

**5. COMPENSATORY STEPPING CORRECTION- BACKWARD**
*Instruction: "Stand with your feet shoulder width apart, arms at your sides. Lean backward against my hands beyond your backward limits. When I let go, do whatever is necessary, including taking a step, to avoid a fall."*
(2) Normal: Recovers independently with a single, large step.
(1) Moderate: More than one step used to recover equilibrium.
(0) Severe: No step, OR would fall if not caught, OR falls spontaneously.

**6. COMPENSATORY STEPPING CORRECTION- LATERAL**
*Instruction: "Stand with your feet together, arms down at your sides. Lean into my hand beyond your sideways limit. When I let go, do whatever is necessary, including taking a step, to avoid a fall."*

*Left*
(2) Normal: Recovers independently with 1 step (crossover or lateral OK).
(1) Moderate: Several steps to recover equilibrium.
(0) Severe: Falls, or cannot step.

*Right*
(2) Normal: Recovers independently with 1 step (crossover or lateral OK).
(1) Moderate: Several steps to recover equilibrium.
(0) Severe: Falls, or cannot step.

**Use the side with the lowest score to calculate sub-score and total score.**

***SENSORY ORIENTATION*** ***SUB SCORE:*** /6

**7. STANCE (FEET TOGETHER); EYES OPEN, FIRM SURFACE**
*Instruction: "Place your hands on your hips. Place your feet together until almost touching. Look straight ahead. Be as stable and still as possible, until I say stop."*
*Time in seconds:_____*
(2) Normal: 30 s.
(1) Moderate: < 30 s.
(0) Severe: Unable.

**8. STANCE (FEET TOGETHER); EYES CLOSED, FOAM SURFACE**
*Instruction: "Step onto the foam. Place your hands on your hips. Place your feet together until almost touching. Be as stable and still as possible, until I say stop. I will start timing when you close your eyes."*
*Time in seconds:_____*
(2) Normal: 30 s.
(1) Moderate: < 30 s.
(0) Severe: Unable.

**9. INCLINE- EYES CLOSED**
*Instruction: "Step onto the incline ramp. Please stand on the incline ramp with your toes toward the top. Place your feet shoulder width apart and have your arms down at your sides. I will start timing when you close your eyes."*
*Time in seconds:_____*
(2) Normal: Stands independently 30 s and aligns with gravity.
(1) Moderate: Stands independently <30 s OR aligns with surface.
(0) Severe: Unable.

***DYNAMIC GAIT*** ***SUB SCORE:*** /10

**10. CHANGE IN GAIT SPEED**
*Instruction: "Begin walking at your normal speed, when I tell you 'fast', walk as fast as you can. When I say 'slow', walk very slowly."*
(2) Normal: Significantly changes walking speed without imbalance.
(1) Moderate: Unable to change walking speed or signs of imbalance.
(0) Severe: Unable to achieve significant change in walking speed AND signs of imbalance.

**11. WALK WITH HEAD TURNS – HORIZONTAL**
*Instruction: "Begin walking at your normal speed, when I say "right", turn your head and look to the right. When I say "left" turn your head and look to the left. Try to keep yourself walking in a straight line."*
(2) Normal: performs head turns with no change in gait speed and good balance.
(1) Moderate: performs head turns with reduction in gait speed.
(0) Severe: performs head turns with imbalance.

**12. WALK WITH PIVOT TURNS**
*Instruction: "Begin walking at your normal speed. When I tell you to 'turn and stop', turn as quickly as you can, face the opposite direction, and stop. After the turn, your feet should be close together."*
(2) Normal: Turns with feet close FAST (≤ 3 steps) with good balance.
(1) Moderate: Turns with feet close SLOW (≥4 steps) with good balance.
(0) Severe: Cannot turn with feet close at any speed without imbalance.

**13. STEP OVER OBSTACLES**
*Instruction: "Begin walking at your normal speed. When you get to the box, step over it, not around it and keep walking."*
(2) Normal: Able to step over box with minimal change of gait speed and with good balance.
(1) Moderate: Steps over box but touches box OR displays cautious behavior by slowing gait.
(0) Severe: Unable to step over box OR steps around box.

**14. TIMED UP & GO WITH DUAL TASK** ***[3 METER WALK]***
*Instruction TUG: "When I say 'Go', stand up from chair, walk at your normal speed across the tape on the floor, turn around, and come back to sit in the chair."*
*Instruction TUG with Dual Task: "Count backwards by threes starting at___. When I say 'Go', stand up from chair, walk at your normal speed across the tape on the floor, turn around, and come back to sit in the chair. Continue counting backwards the entire time."*

*TUG:_____seconds; Dual Task TUG:_____seconds*
(2) Normal: No noticeable change in sitting, standing or walking while backward counting when compared to TUG without Dual Task.
(1) Moderate: Dual Task affects either counting OR walking (>10%) when compared to the TUG without Dual Task.
(0) Severe: Stops counting while walking OR stops walking while counting.

**When scoring item 14, if subject's gait speed slows more than 10% between the TUG without and with a Dual Task the score should be decreased by a point.**

**TOTAL SCORE:** ____/28

https://www.sralab.org/sites/default/files/2017-06/MiniBEST_revised_final_3_8_13.pdf

# Additional Reading

- Shumway-Cook Anne & Woollacott MH. Clinical Management of the patient with a postural control disorder. In Motor Control: Translating Research into Clinical Practice 4th Edition. Chapter 7